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Effects of Content and Language Integrated Learning on secondary-level students' content learning: A meta-analysis

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Abstract

Content and Language Integrated Learning (CLIL) has demonstrated benefits for target language development, specifically because the target language is used as the medium of instruction. However, existing primary studies report conflicting results regarding CLIL's effects on knowledge acquisition of non-language content subjects. This lack of consistency in findings suggests that synthesis of CLIL study results on content subjects is necessary. To fill this gap, the present study identified 21 between-group CLIL studies on the acquisition of content knowledge among secondary school students over the past two decades, and used a multilevel meta-analysis to estimate the overall effectiveness of the CLIL approach. Based on 29 independent samples and 39 effect sizes (N = 36,905), CLIL's general effectiveness regarding secondary-level students' knowledge gains in content subjects was 0.09 (SE = 0.10, z = 0.85, p = .40, 95% CI [−0.11, 0.28]), indicating comparable gains of the CLIL approach with that of its mainstream counterpart. The results of the moderator analyses further confirmed the homogeneity of CLIL and mainstream groups and the nature of the content subjects as significant variables that may moderate the effectiveness of the CLIL approach.

Keywords: Content/language integrated learning; content learning; English medium instruction; Meta-analysis; Secondary school

1. Introduction

Content and Language Integrated Learning (CLIL) is a pedagogical approach where students develop proficiency in the target language (TL), typically English, by learning content subjects (e.g., physics, mathematics, history) in that language (Coyle et al., 2010). CLIL has received

much attention in European countries (e.g., Canga Alonso & Arribas Garcia, 2015; Fernandez-Sanjurjo et al., 2019; Martinez Agudo, 2020; Rumlich, 2018; San Isidro & Lasagabaster, 2019) and is considered pedagogically useful in that it "seeks to develop competence in both the non-language subject and the language in which it is taught, giving equal importance to each" (European Commission, 2006, p. 7). However, unlike the well-balanced expectations for CLIL programs, a non-proportionate amount of attention within the CLIL research community has been directed especially towards studying their impact on the growth of TL proficiency rather than that of content knowledge. This trend has been observed in research with secondary school students (e.g., Goris et al., 2013; Martinez Agudo, 2020; San Isidro & Lasagabaster, 2019; Yang, 2015) and primary ones (e.g., Canga Alonso, 2015; Jafarigohar et al., 2022; Pladevall-Ballester & Vallbona, 2016). Nevertheless, the extent to which CLIL programs can enhance secondary-level students' TL development is conclusive, indicating that they are generally beneficial. For example, a recent meta-analysis by Lee et al. (2023) involving 44 samples from 38 primary studies (N = 7434) focused on examining the effect of CLIL on secondary school students' TL development; the average effect size for the short term was 0.73, and that for the long term was 1.01.

However, unlike TL developments, the existing literature has presented somewhat contradictory results depending on the context and other aspects of the intervention. For example, some studies (e.g., Binterová et al., 2014; Ouazizi, 2016) have reported some student gains in content knowledge achieved through CLIL instruction, whereas others (e.g., Fernández-Sanjurjo et al., 2019; Piesche et al., 2016; Virdia, 2022) have shown that CLIL students did not achieve results comparable to those of students in mainstream conditions, thus suggesting the potential negative effects of the approach. Therefore, the extent to which CLIL programs can enhance students' learning of (non-language) content subjects has remained inconclusive. Thus, despite the aforementioned improvements in language skills, the uncertain effectiveness of CLIL programs raises a legitimate concern that students in CLIL learning conditions may fall behind those in content learning areas. Given this concern, we cannot deny the possibility that CLIL programs may prioritize language skills at the expense of content knowledge acquisition (Macaro, 2018; Ruiz-Cecilia et al., 2023).

The field has indeed witnessed some synthetic efforts to explore the effectiveness of CLIL for content learning (e.g., Graham et al., 2018; Hidalgo & Ortega-Sanchez, 2023; Murphy et al., 2020; Ruiz-Cecilia et al., 2023), all of which have contributed valuable insights to resolve the abovementioned concerns. However, most of these research syntheses are rather limited because of their descriptive and qualitative nature (e.g., Graham et al., 2018; Hidalgo & Ortega-Sanchez, 2023; Ruiz-Cecilia et al., 2023). Three full-scale meta-analyses

(Cimermanová, 2020; Lee et al., 2023; Lo & Lo, 2014) have limited implications, as Lee et al. and Cimermanová only considered TL development (rather than content learning), and Lo and Lo lacked comprehensiveness, focusing exclusively on studies conducted in Hong Kong.

Acknowledging these limitations, the present study aims to assess the overall effectiveness of the CLIL approach for improving secondary-level students' content learning outcomes. It focuses on secondary-level students for the following reasons. First, much of the existing CLIL empirical research has focused on secondary school students, which provides enough samples and effect sizes for synthesizing research results with a meta-analytical approach. Second, secondary schools offer more content subjects in different key learning areas (e.g., science, humanities, social sciences), often with high-stakes examination. This provides more data to evaluate the effectiveness of CLIL in student content learning and allows potential comparison of different content subjects. Additionally, this study aims to identify a list of potential variables related to this effectiveness and their moderating effects, given the contextual and methodological differences across extant CLIL studies (see Hidalgo and Ortega-Sanchez (2023) for the importance of contextual factors when considering the effects of CLIL on content knowledge). Finally, the present meta-analysis attempts to widen its scope by considering empirical studies under the label "English Medium Instruction (EMI)," but focusing on secondary-level students because of their functional equivalence to CLIL (Rose et al., 2021). Additionally, including EMI studies enables us to align our findings with those of Lee et al.'s (2023) meta-analysis on TL development, which included both CLIL and EMI studies. The results of Lee et al.'s study and those of the current meta-analysis are, thus, expected to contribute to the discussion regarding the cost-benefit ratio for content and TL learning (Macaro, 2018).

2. Literature review

2.1. Theoretical underpinnings behind CLIL and challenges in reality

CLIL is grounded in Second Language Acquisition (SLA) theories (Gabillon, 2020), which underscore the importance of TL input, interaction, and output for students' language development. In addition to the amount of language input and output, the authentic nature of the input provided by content subjects and the meaningful tasks students engage in are arguably more effective for TL learning (e.g., Gallardo-del-Puerto et al., 2020). These benefits of CLIL on TL learning have gathered some support from empirical studies, as summarized by Lee et al.'s (2023) meta-analysis. Comparatively, content learning in CLIL, which can be operationalized as students' academic achievement in content subjects, has been less discussed or theorized. This is intriguing, as content learning is one of the two (theoretically) equal goals

of CLIL. One possible reason for this is that most CLIL research has been conducted by applied linguists, thereby lacking input from content specialists (Dalton-Puffer et al., 2022). Two theoretical frameworks shed some light on the discussion.

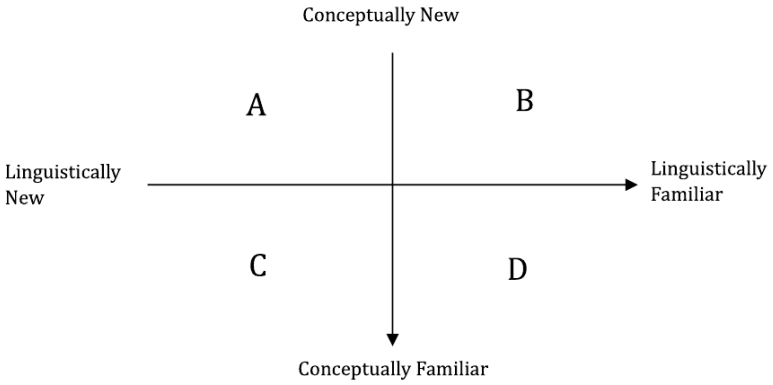


Fig. 1. Relationships between the conceptual and linguistic development of immersion students (Adapted from Johnson & Swain, 1994, p. 212).

The first is Cummins' (2000, 2021), theoretical model of bilingual education. This model postulates that students' educational outcomes are the function of the interaction of background, students' input, and educational programs. In particular, the Developmental Interdependence Hypothesis and the Threshold Hypothesis incorporated into the model highlight the importance of attaining a threshold level of first language (L1) and TL competence when intensive exposure to TL begins; thus, students can develop their TL without sacrificing academic achievement. In this sense, when secondary-level students start their CLIL experience, assuming they have attained a certain level of L1 and TL proficiency, they can enjoy the benefits of CLIL.

Johnson and Swain's (1994) framework (originally designed for immersion students, but also applicable for CLIL) similarly juxtaposes the relationship between students' language and conceptual development. This framework (Fig. 1) hypothesizes that when students start immersion or CLIL at a later educational stage (e.g., senior primary or secondary level), they have already learned some academic concepts through their L1 and may then rely on L1 internally for processing academic content (Benson, 2004). In this case, they will be in Segment C in Fig. 1, where they must bridge the proficiency gap between their TL proficiency and knowledge of academic concepts. Furthermore, content knowledge often becomes more abstract and difficult at senior grade levels. Hence, students who start immersion or CLIL later may encounter the challenges in Segment A, where they must master both academic concepts and TL simultaneously. It has also been argued that the TL required to succeed in CLIL or other bilingual education programs comprises different registers, including language for

classroom interaction (i.e., everyday language) and language associated with disciplines (i. e., academic language or academic literacies) (Lo & Lin, 2019). The latter has been shown to be complicated, with a high density of subject-specific vocabulary and specific sentence patterns and genres (Martin, 2013) that create challenges for students to express their content knowledge in the TL. For instance, Lo et al. (2023) analyzed the biology public examination results of EMI students in Hong Kong and showed that their performance was negatively affected when asked to answer questions in sentences or texts.

The above discussion simultaneously illustrates the potential benefits of CLIL and highlights the challenges students may encounter, particularly in the domain of content learning. It is necessary to provide more conclusive evidence regarding the effectiveness of CLIL.

2.2. Potential moderator variables

Based on CLIL literature, we identified a set of potential moderator variables that could influence the effectiveness of the CLIL approach. These moderators were (1) homogeneity confirmation, (2) the nature of the content subject, (3) the test language, and (4) the nature of the CLIL implementation. In the following discussions, we review the relevant literature and the rationale for selecting each moderator.

2.2.1. Homogeneity confirmation

Research on the relative effects of CLIL and mainstream instruction has been criticized for neglecting to assess the prior knowledge and competence of participating groups (e.g., Bruton, 2011a, 2011b; Graham et al., 2018); thus, the positive effects of CLIL on students' learning outcomes, if any, cannot be attributed solely to instruction. That is, given that "CLIL can attract a disproportionally large number of academically bright students" (Mehisto, 2007, p. 63), the superiority of CLIL should be, at least partially, attributed to pre-existing differences between CLIL and mainstream (i.e., comparison) groups. In response to such concerns, more recent studies have tried to guarantee the homogeneity of CLIL and non-CLIL groups by conducting a pre-test on the target content knowledge (e.g., Fung, 2021; Ohlberger & Wegner, 2017; Rosi, 2018); they found no statistical difference between the two groups. Some studies have measured different cognitive and affective aspects (e.g., verbal intelligence, TL proficiency, and motivation), and matched the two groups in terms of these aspects (e.g., Hughes & Madrid, 2020; Martinez Agudo, 2022; Pascual Bajo, 2018; Pérez-Cañado, 2018). Given such methodology of recent CLIL studies, Pérez-Cañado (2023) suggests that "the differences detected between homogenous CLIL and non-CLIL groups can be ascribed predominantly to CLIL" (p. 362).

In a meta-analysis by Lo and Lo (2014), the significance of controlling for prior knowledge of EMI students was emphasized, and the homogeneity of CLIL and mainstream groups was confirmed as a significant moderator of secondary-level Hong Kong students' academic achievements. Similarly, in the meta-analysis by Lee et al. (2023), the confirmation of the homogeneity of the two groups was found to be a significant moderator for the development of secondary-level students' English proficiency. Accordingly, in alignment with these studies, homogeneity confirmation was considered a moderator in the current study.

2.2.2. Nature of the content subject

CLIL research has examined instruction in social sciences and humanities, which involves a lot of language, and in mathematics and science (e.g., Lau & Yuen, 2011; Mulder, 2016; Ouazizi, 2016), which involves less language. In social studies and humanities lessons, teachers engage students in interactions and question-answer sequences that elicit students' personal experiences and opinions (Lo, 2014). However, in mathematics and science lessons, teachers and students often mobilize other resources, such as symbolic devices, manipulatives (e.g., pictures, charts, and graphs), hands-on experiments, and teachers' demonstrations, to explain the abstract concepts involved (O'Halloran, 1998). Given these differences, the nature of the target subject may affect the nature of language use and instructional practice in CLIL lessons (e.g., Lo, 2019; Villabona & Cenoz, 2022). However, few synthetic efforts have been conducted to estimate the effects of the CLIL approach on the acquisition of knowledge in different content subjects (Murphy et al., 2020).

2.2.3. Test language

When evaluating the outcomes of content knowledge, questions arise regarding the language in which students should be tested. This is particularly important when evaluating the effectiveness of bilingual programs (Canz et al., 2021) like CLIL. Indeed, whether students' L1 should be used to test their content knowledge has been an important issue in CLIL assessments (Sato, 2023), given that the instruction and subject-specific terms are generally delivered in the TL; that is, the CLIL group could be disadvantaged when tested in their L1 (Coyle et al., 2010). To date, some studies have employed the students' L1 for assessment (e.g., Anghel et al., 2016; Surmont et al., 2016), whereas others have employed assessments in the TL (e.g., Galera Rodriguez, 2022; Lau & Yuen, 2011; Ohlberger & Wegner, 2017), or even a mixture of the two (e.g., Piesche et al., 2016; Xanthou, 2011), with the latter studies considering the close connection between the TL, encoding knowledge structures, and processing knowledge (Wolff & Holmes, 2011). Taking into account the education level of the target population in this meta-analysis (i.e., secondary school students), it can be suggested that they may be less influenced by test-language effects (Canz et al., 2021) than their lower-

level counterparts; however, no systematic investigation has been conducted into this moderator in any meta-analysis on CLIL. Therefore, the language of the test was considered a moderator.

2.2.4. Nature of CLIL implementation

Another variation in CLIL research has been influenced by the nature of the CLIL approach administered at sampled schools (Lo & Lo, 2014). Most studies have been conducted in schools in which the CLIL curriculum has been implemented for years, such as in Hong Kong (e.g., Lau & Yuen, 2011; Salili & Lai, 2003) and Spain (Hughes & Madrid, 2020; Martinez Agudo, 2022; Pérez-Cañado, 2018). In some studies (e.g., Meyerhoffer & Dreesmann, 2019; Piesche et al., 2016), the CLIL approach was introduced as a new intervention for students having little or no previous experience with such a bilingual approach. It is possible, then, that the nature of the CLIL implementation (i.e., as a novel intervention or as part of a regular curriculum) may influence its effectiveness, with this approach being less disadvantageous (in terms of content learning) for learners accustomed to such instruction (e.g., Graham et al., 2018; Hughes & Madrid, 2020; see a similar point regarding academic growth in mathematics in Ruiz-Cecilia et al. (2023). In other words, studies employing CLIL as a short-term intervention may not effectively capture the effects of this bilingual program (e.g., Lo & Lo, 2014; Virdia, 2022). Therefore, the nature of the CLIL approach at the sampled institutions was included as a potential moderator.

3. Present study

This meta-analysis examined the effects of CLIL instruction on the acquisition of content knowledge compared to its mainstream counterpart for secondary-level students, given the lack of quantitative synthetic efforts in this regard. Generally, previous meta-analyses primarily focused on determining whether a specific intervention (i.e., treatment) yields superior learning outcomes compared with control or comparison conditions. However, considering the ongoing discussions in the CLIL literature, this study did not aim to establish the superiority of CLIL instruction with regard to students' content learning. Instead, it endeavored to investigate whether CLIL participants experienced any disadvantages in content learning given the increased challenge of learning content in TL. Therefore, we sought to answer the following two research questions:

1. To what extent are secondary-level students receiving CLIL instruction disadvantaged in acquiring content knowledge compared to those receiving mainstream instruction?
2. To what extent are the identified moderators related to the effects of CLIL on secondary-level students' acquisition of content knowledge?

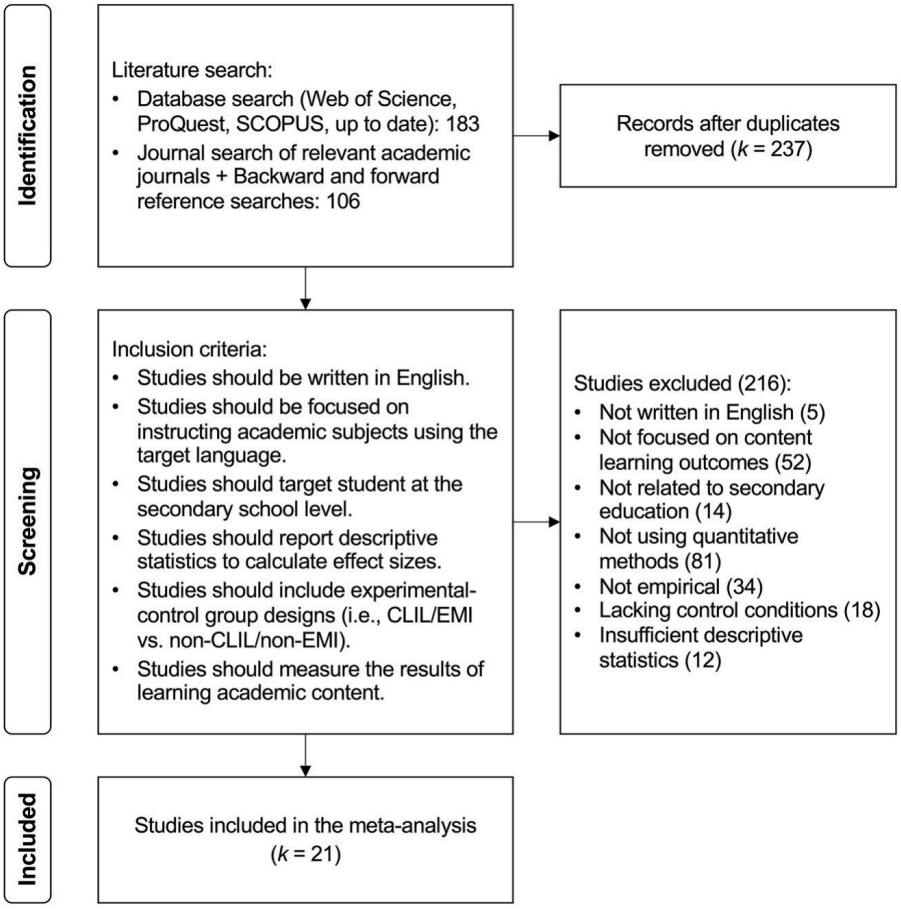


Fig. 2. PRISMA flow diagram of the literature search.

4. Methods

4.1. Literature search and inclusion criteria

Based on the literature search, the inclusion criteria for reviewing primary data were identified as follows: studies were required to a) be written in English; b) focus on providing instruction on academic subjects using the TL; c) target students at the secondary school level; d) report sufficient descriptive statistics (i.e., sample size of each group, mean, and standard deviations) for effect size calculation; e) employ experimental control group designs (e.g., CLIL/EMI versus non-CLIL/non-EMI groups); and f) measure the results of learning academic content. The process employed for the literature search and the inclusion of studies in the meta-analysis is demonstrated in the PRISMA flow diagram in Fig. 2.

Table 1
List of included samples and their sample sizes.

Independent sample	Title of the article	Sample size	
		CLIL group	Mainstream group
Dallinger et al. (2016)	The effect of content and language integrated learning on students' English and history competences–Killing two birds with one stone?	483	444
Fung (2021)	Teaching science through home and second languages as the medium of instruction: A comparative analysis of junior secondary science classrooms in Hong Kong	239	240
Galera Rodríguez (2022) [1]	Analysing content acquisition in CLIL vs. non-CLIL programmes	14	14
Galera Rodríguez (2022) [2]		18	14
Gregorczyk (2012)	An empirical study on the acquisition of content in a CLIL-based chemistry course: A preliminary report	31	245
Ho and Man (2007)	Student performance in Chinese medium-of instruction (CMI) and English medium-of-instruction (EMI) schools: What we learned from the PISA study	724	2702
Hughes and Madrid (2020)	The effects of CLIL on content knowledge in monolingual contexts	65	74
Lau and Yuen (2011) [a] [1]	The impact of the medium of instruction: The case of teaching and learning of computer programming	84	25
Lau and Yuen (2011) [2]		53	44
Madrid and Barrios (2018)	A comparison of students' educational achievement across programmes and school types with and without CLIL provision	102	101
Martínez Agudo (2022)	To what extent does parental educational background affect CLIL learners' content subject learning? Evidence from research in Spain	74	82
Meyerhoffer and Dreesmann (2019)	The exclusive language of science? Comparing knowledge gains and motivation in English-bilingual biology lessons between non-selected and preselected classes	85	73
Mulder (2016)	Learning biology in Dutch bilingual education: Any reason for concern?	16	49
Oattes et al. (2022) [1]	A showdown between bilingual and mainstream education: the impact of language of instruction on learning subject content knowledge	324	176
Oattes et al. (2022) [2]		336	291
Ohlberger and Wegner (2017)	Measuring the knowledge increase of eight grade students in a bilingual biology unit	25	19
Ouazizi (2016)	The effects of CLIL education on the subject matter (mathematics) and the target language	10	20
Pascual Bajo (2018)	The impact of content and language integrated learning (CLIL) on Valencian secondary education students' English language acquisition and academic achievement: A case study	21	25
Pérez-Cañado (2018)	The effects of CLIL on L1 and content learning: Updated empirical evidence from monolingual contexts	406	422

Independent sample	Title of the article	Sample size	
		CLIL group	Mainstream group
Piesche et al. (2016)	CLIL for all? A randomized controlled field experiment with sixth-grade students on the effects of content and language integrated science learning	362	360
Rosi (2018)	Content-specific learning in CLIL: The case of physics teaching in Italy	16	18
Salili and Lai (2003) [1]	Learning and motivation of Chinese students in Hong Kong: A longitudinal study of contextual influences on students' achievement orientation and performance	95	316
Salili and Lai (2003) [2]		235	166
Salili and Lai (2003) [3]		368	316
Salili and Lai (2003) [4]		212	166
Surmont et al. (2016)	The effects of CLIL on mathematical content learning: A longitudinal study	35	72
Yip et al. (2003) [1]	Evaluation of the effects of medium of instruction on the science learning of Hong Kong secondary students: Performance on the science achievement test	4624	3630
Yip et al. (2003) [2]		4624	4552
Yip et al. (2003) [3]		4624	3969

[a] Lau and Yuen (2011) included three different bands, and Band 1 was not included in the present meta-analysis, as the number of students in the CLIL group was only one.

The literature search in databases, such as ProQuest, Scopus, and Web of Sciences, was conducted using the following keywords: ["English as a medium instruction" OR EMI OR "Content and language integrated learning" OR CLIL] AND [outcome(s) OR development OR effect(s)]. During this stage of the literature search, 183 studies conducted over the past two decades were identified. To conduct backwards and forwards searches, the reference lists of previously recognized studies, systematic reviews (Graham et al., 2018; Hidalgo & Ortega-Sanchez, 2023; Murphy et al., 2020; Ruiz-Cecilia et al., 2023), and one meta-analysis (Lo & Lo, 2014) were examined; 106 studies were identified. Finally, 237 studies remained after removing duplicates among the studies found at this stage.

Next, 237 studies were reviewed to determine whether they satisfied the aforementioned criteria. Accordingly, based on the inclusion criteria, 216 studies were eliminated: a) five were not written in the English language, b) 52 did not focus on content learning outcomes, c) 14 were not related to secondary education contexts, d) 81 were not using quantitative methods (i.e., qualitative studies), e) 34 were not empirical studies (e.g., literature reviews, systematic reviews, meta-analysis), f) 18 were lacking control conditions, and g) 12 did not include the sufficient descriptive statistics required for effect size calculation (see Supplementary Material

for a list of exclusion criteria and excluded studies). Notably, authors of studies with insufficient descriptive statistics were contacted; however, the required materials were not provided. Finally, 21 studies were included in the meta-analysis.

4.2. Dataset construction

The final dataset included 29 samples (N = 36,905) from 21 studies (see Table 1), as some studies had more than one sample according to the following two patterns. First, they included multiple separate groups of students. For example, Lau and Yuen (2011) examined students in two different bands and obtained two independent samples. Oattes et al. (2022) simultaneously investigated students in two different grades (Grades 7 and 9); this resulted in two independent studies. Second, the studies had more than one comparison. For example, Galera Rodriguez (2022) had two experimental groups varying in the intensity of the CLIL program (i.e., hard and soft CLIL) and one control group; this design led us to compute two comparisons. Yip et al. (2003) included one CLIL group and three control groups, resulting in three comparisons. Salili and Lai (2003) included four different comparisons, which resulted from two comparisons based on two different EMI groups and one control group; these were further split based on gender.

4.3. Effect size calculation

A standardized measure of difference was needed to compare the effectiveness of CLIL programs with mainstream approaches. For this purpose, we chose Hedges' g (Hedges, 1981), a type of effect size unit designed to express the size of the effect of an intervention. Hedges' g is particularly useful because it accounts for variations in sample sizes across studies, making our comparisons more reliable, especially when dealing with smaller groups of participants (e.g., N < 50). This measure was calculated by examining the difference between the average post-test scores of the CLIL group and those of the mainstream education group. These differences were then scaled according to the combined standard deviation of both groups; this helped to normalize the results across different studies. To ensure the accuracy of our findings, Hedges' g includes a correction factor that adjusts for bias, which is crucial when comparing studies with varying numbers of participants.

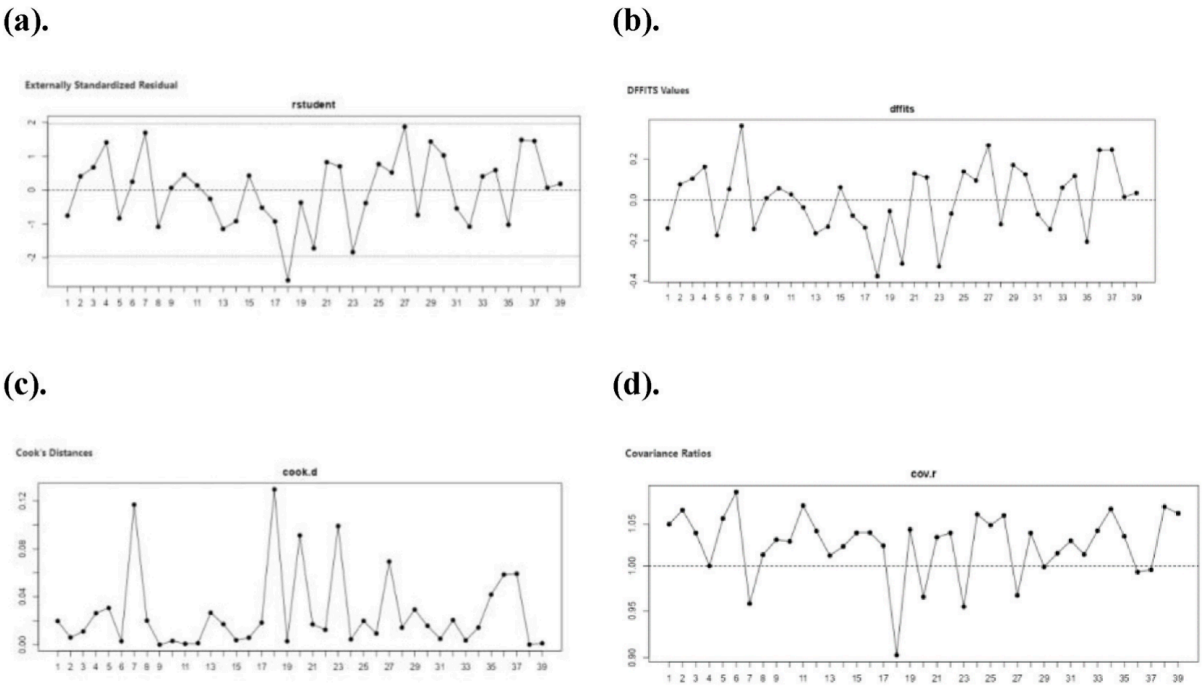


Fig. 3. Plot of the (a) studentized deleted residuals, (b) DFFITS values, (c) Cook's distances, and (d) COVRATIO values for 39 effect sizes of post-tests for the comparison of CLIL versus mainstream conditions.

In instances in which studies presented multiple measurements, such as that conducted by Rosi (2018), we faced the challenge of overlapping data points, which could potentially skew the overall analysis. To circumvent the issue of statistical dependence, we employed a multilevel meta-analysis approach (see *Data Analysis Plan* section for details). This technique allowed us to incorporate multiple data points from the same study by analyzing them within their specific contexts, thus maintaining the integrity of our overall effect size calculation. This approach was instrumental in allowing us to aggregate 39 post-test effect sizes from different studies, providing a comprehensive and unbiased view of the impact of CLIL programs compared to mainstream learning conditions.

4.4. Outlier diagnosis and publication bias

The statistical software Jamovi (version 2.4.8) (The Jamovi Project, 2023) was used to detect outliers within the calculated effect size dataset. We checked outlier indicators, such as studentized deleted residuals, difference in fit(s) (DFFITS) values, Cook's distances, and covariance ratio (COVRATIO) values, with outliers represented by a red circle if extant. The examination concluded that no outliers were present in the dataset, thereby confirming the consistency of the studies included in the analysis (Fig. 3).

We further checked for publication bias by considering both a funnel plot and Egger's regression analysis (Egger et al., 1997) by employing a software program, STATA (version 16.1) (StataCorp, 2020). The funnel plot in Fig. 4 reveals a symmetrical pattern. Given the subjective nature of assessing publication bias using the funnel plot, Egger's regression analysis was performed to scrutinize any potential publication bias. The findings indicate no significant bias ($z = 0.40$, $p = .69$).

4.5. Coding moderator variables

To check what values the independent samples reported for the moderator variables, such as (1) homogeneity confirmation, (2) the nature of the content subject, (3) the language of the test, and (4) the nature of the CLIL implementation, the authors independently reviewed relevant information from the included studies, with an agreement coefficient of 0.92. Disagreements were resolved by reaching consensus. The values of the moderator variables for each independent sample are described in Table 2.

First, regarding "homogeneity of confirmation," the independent samples included in the current meta-analysis were categorized as "confirmed" if they a) involved conducting a pre-test on the target content knowledge and demonstrated no statistical difference between the two groups; b) measured aspects, such as verbal intelligence, TL proficiency, and motivation, and matched the two groups in terms of these aspects; and c) implemented a randomized sampling method. Conversely, samples were categorized as "not confirmed" when a) they had no indication about the homogeneity of the CLIL and mainstream groups, and b) the groups differed significantly based on pre-test results.

Second, for the "nature of the content subject," the samples were categorized into two types, given the potential variations in cognitive load resulting from the amount of TL interaction required of the subject being instructed: language more loaded (e.g., social sciences and humanities), and language less loaded (e.g., mathematics, computer sciences, and natural sciences).

Third, regarding the "nature of CLIL implementation," the samples were categorized as "intervention" when involving participants with no prior CLIL experience, or "curriculum" when CLIL was integrated into the existing curriculum at the sampled institution.

Finally, for "test language," the samples were categorized as "in target language" when the TL was included as the language for assessment in the CLIL group, and as "others" when the assessment was administered in the participants' L1 or when the post-test data were based on the participants' end-of-the-semester grade or equivalents. In one study (Oattes et al., 2022),

the CLIL group was administered a post-test consisting of a section in the participants' L1 and another in the TL. Thus, the scores for each section were coded differently based on this moderator.

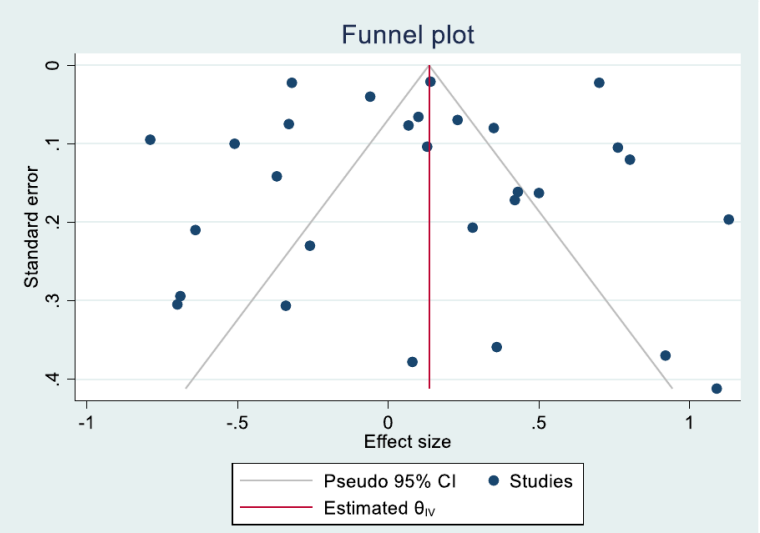


Fig. 4. Funnel plot for post-test effect sizes.

Table 2
Moderator variables and their values for the independent samples.

Independent sample	Homogeneity confirmation	Nature of the Content Subject	Nature of CLIL Implementation	Test Language
Dallinger et al. (2016)	Confirmed	More loaded	Curriculum	Others[a]
Fung (2021)	Confirmed	Less loaded	Intervention	In target language
Galera Rodríguez (2022) [1]	Not confirmed	More loaded	Curriculum	Others
Galera Rodríguez (2022) [2]	Not confirmed	More loaded	Curriculum	Others
Gregorczyk (2012)	Not confirmed	Less loaded	Intervention	Others
Ho and Man (2007)	Not confirmed	Less loaded	Curriculum	Others
Hughes and Madrid (2020)	Confirmed	Less loaded	Curriculum	Others
Lau and Yuen (2011) [1]	Confirmed	Less loaded	Curriculum	In target language
Lau and Yuen (2011) [2]	Confirmed	Less loaded	Curriculum	In target language
Madrid and Barrios (2018)	Confirmed	Less loaded	Curriculum	Others
Martinez Agudo (2022)	Confirmed	More loaded	Curriculum	Others
Meyerhoffer and Dreesmann (2019)	Not confirmed	More loaded	Intervention	Others

Independent sample	Homogeneity confirmation	Nature of the Content Subject	Nature of CLIL Implementation	Test Language
Mulder (2016)	Not confirmed	More loaded	Curriculum	In target language
Oattes et al. (2022) [1]	Not confirmed	More loaded	Curriculum	In target language & Other language (each for different test sections)
Oattes et al. (2022) [2]	Not confirmed	More loaded	Curriculum	In target language & Other language (each for different test sections)
Ohlberger and Wegner (2017)	Confirmed	More loaded	Intervention	In target language
Ouazizi (2016)	Not confirmed	Less loaded	Curriculum	Others
Pascual Bajo (2018)	Confirmed	More loaded	Curriculum	Others
Pérez-Cañado (2018)	Confirmed	Less loaded	Curriculum	In target language
Piesche et al. (2016)	Confirmed	Less loaded	Intervention	In target language
Rosi (2018)	Confirmed	Less loaded	Intervention	Others
Salili and Lai (2003) [1]	Not confirmed	Less loaded	Curriculum	In target language
Salili and Lai (2003) [2]	Not confirmed	Less loaded	Curriculum	In target language
Salili and Lai (2003) [3]	Not confirmed	Less loaded	Curriculum	In target language
Salili and Lai (2003) [4]	Not confirmed	Less loaded	Curriculum	In target language
Surmont et al. (2016)	Confirmed	Less loaded	Intervention	Others
Yip et al. (2003) [1]	Not confirmed	Less loaded	Curriculum	In target language
Yip et al. (2003) [2]	Not confirmed	Less loaded	Curriculum	In target language
Yip et al. (2003) [3]	Not confirmed	Less loaded	Curriculum	In target language

[a] Participants' scores at Grade 8-T1 were used for the analysis.

4.6. Data analysis plan

For data analysis, we used STATA (version 16.1) (StataCorp, 2020). Initially, we determined the overall average effect size by synthesizing the calculated 39 effect sizes across 29 samples. Next, we examined the impact of specific moderators on the efficacy of the CLIL programs. As mentioned in the section *Effect size calculation*, the effect sizes were computed at the measurement level, where they were nested within each sample, leading us to conduct a two-level meta-analysis. This analysis was performed using the *meglm* command in STATA, creating multilevel mixed-effects generalized linear models based on the effect sizes. Given that effect sizes come with estimated values and standard errors, it was necessary to use the variance of these sizes instead of within-group variability in our models to properly reflect the structure of the datasets.

By constructing these models, we aimed to achieve a secondary goal: to investigate the moderation effects by incorporating moderator variables as independent variables. This process began with four simple regression analyses, each treating a different moderator as the sole independent variable to determine its statistical significance. If significant, these moderators were collectively included as independent variables in a subsequent analysis to ascertain if their significance persisted. This advanced meta-regression model allowed us to assess the moderation effects through the coefficients of the moderator variables, presuming other variables in the model remained constant.

5. Results

5.1. The overall effectiveness of CLIL for improving students' content learning

For general interpretation purposes, a forest plot is shown in Fig. 5 to provide an overview of the post-test effect sizes calculated for each sample (k = 29). Among 29 samples, the average effect sizes ranged from −0.79 (Fung, 2021) to 1.13 (Gregorczyk, 2012), indicating variations in the findings regarding CLIL's effectiveness for secondary-level students' content learning. Furthermore, nine samples (i.e., Dallinger et al., 2016; Galera Rodriguez, 2022 [1]; Galera Rodriguez, 2022 [2]; Ho & Man, 2007; Lau & Yuen, 2011 [1]; Ohlberger & Wegner, 2017; Salili & Lai, 2003 [3]; Salili & Lai, 2003 [4]; Surmont et al., 2016) featured zeros in the 95% confidence intervals of their effect sizes, indicating that these samples revealed the null intervention effects of CLIL instruction for students' content learning.

To compute the overall average effect size estimate by synthesizing the calculated 39 effect sizes, we built a multilevel regression model as a variance-known version (i.e., using standard errors of effect sizes as within-group variability). The results are shown in Table 3. The overall average effect size was 0.09 (SE = 0.10, z = 0.85), which was not statistically significant (p = .40, 95% CI [−0.11, 0.28]). This indicates that the overall effect of CLIL programs on secondary-level students' knowledge gains in content subjects did not differ from that of mainstream instruction.

5.2. Moderator analyses

After examining overall effectiveness, we investigated the extent to which the identified moderators influenced overall effectiveness. As detailed in the *Data analysis plan* section, we first conducted simple meta-regression analyses (without considering other moderators) to examine how each of the identified moderators was related to the effectiveness of the CLIL approach. As can be seen in Table 4, the results of simple meta-regression analyses revealed that "Homogeneity confirmation" made statistically significant contributions to the overall

effectiveness at the 5% significance level, and “Nature of the content subject” and “Test language” did so at the 10% significance level, whereas “Nature of CLIL Implementation” did not.

Considering this result, “Homogeneity confirmation” and “Nature of the content subject” and “Test language” were further included in the multiple meta-regression in order to compute more precise coefficients (see Table 5). The results showed that the effectiveness of CLIL was diminished in situations where (1) baseline differences between CLIL and mainstream conditions were confirmed (β = −0.54, *SE* = 0.17, *z* = −3.14, *p* = .002, 95% CI [−0.87, −0.20]) and (2) the target content subjects required a higher degree of language involvement (β = −0.61, *SE* = 0.24, *z* = −2.58, *p* = .01, 95% CI [−1.08, −0.15]).

6. Discussion

In the CLIL literature, the effects of the CLIL approach on the acquisition of (non-language) content subjects have remained largely inconclusive (e.g., Dalton-Puffer, 2011; Hughes & Madrid, 2020; Rosi, 2018), with existing primary studies showing contrasting results (e.g., Binterova et al., 2014; Fung & Yip, 2014; Ouazizi, 2016; Piesche et al., 2016). The present study was conducted to provide some quantitative and synthetic evidence on this issue.

The results of this meta-analysis, based on 39 effect sizes from 29 samples, revealed that the overall effect size of CLIL, in comparison to the mainstream, in terms of the acquisition of content knowledge was 0.09, with no significant difference, indicating that the two conditions produced statistically comparable gains. Therefore, the synthesized results of the present meta-analysis indicate no negative outcomes in content learning resulting from the implementation of the CLIL approach. Considering Cummins’ (2000, 2021) theoretical model of bilingual education, secondary-level students of the identified studies could be assumed to have attained a certain threshold level of L1 and TL proficiency, to the extent that their content learning would not be disadvantaged when exposed to CLIL instruction.

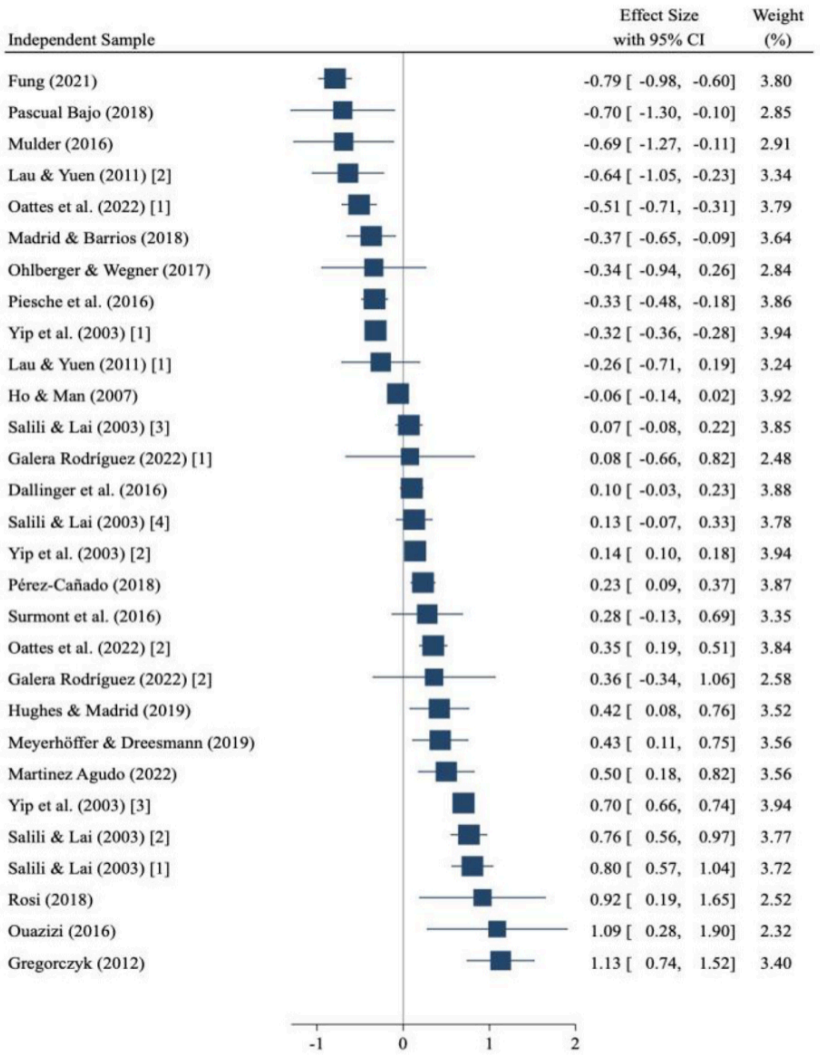


Fig. 5. A forest plot for post-test effect sizes.

Table 3
Overall mean effect size (ES) estimates for CLIL instruction for content learning.

No. of ESs	No. of Sample	Mean ES	SE	95% CI
39	29	0.09	0.10	−0.11 ~ 0.28

Table 4
Simple meta-regression for each moderator variable.

Moderator variable	Category	# of ES	Mean ES	Post-test (*k* = 29)				
				Est.	*SE*	*z*	*p*	95% CI
Homogeneity confirmation	Confirmed	20	−0.16	−0.44[a]	0.21	−2.14	0.03	−0.85 ~ −0.04
	Not confirmed	19	0.28	(reference)				
Nature of the content subject	Language (more loaded)	12	−0.23	−0.43[b]	0.25	−1.70	0.09	−0.92 ~ 0.06
	Language (less loaded)	27	0.19	(reference)				

Moderator variable	Category	# of ES	Mean ES	Post-test (*k* = 29)				
				Est.	*SE*	*z*	*p*	95% CI
Test language	In target language	24	−0.09	−0.39[b]	0.21	−1.81	0.07	−0.81 ~ 0.03
	Others	15	0.30	(reference)				
Nature of CLIL Implementation	Curriculum	31	0.13	−0.08	0.25	−0.32	0.75	−0.58 ~ 0.42
	Intervention	8	0.05	(reference)				

Note. The reference level of each moderator variable is marked "(reference)". Learners' ages and whether the study was labelled as CLIL or EMI were statistically controlled to estimate the values more accurately.

[a] $p < .05$.

[b] $p < .10$.

Table 5

Multiple meta-regression for each moderator variable.

Moderator variable	Category	# of ES	Post-test (*k* = 29)				
			Est.	*SE*	*z*	*p*	95% CI
Homogeneity confirmation	Confirmed	20	−0.54[b]	0.17	−3.14	0.002	−0.87 ~ −0.20
	Not confirmed	19	(reference)				
Nature of the content subject	Language (more loaded)	12	−0.61[a]	0.24	−2.58	0.01	−1.08 ~ −0.15
	Language (less loaded)	27	(reference)				
Test language	In target language	24	−0.26	0.19	−1.41	0.16	−0.63 ~ 0.10
	Others	15	(reference)				

Note. The reference level of each moderator variable is marked "(reference)". Learners' ages and whether the study was labelled as CLIL or EMI were statistically controlled to estimate the values more accurately.

[a] $p < .05$.

[b] $p < .01$.

It is worth discussing this finding along with that of Lee et al.'s (2023) meta-analysis of the effect of CLIL on improving secondary-level students' TL competence; this research revealed a medium-sized overall effect. Taken together, Lee et al.'s findings and those of the present meta-analysis indicate the positive and comparable effects of CLIL on secondary-level students' TL development and content learning compared to mainstream conditions, respectively. These findings collectively attest to the underlying proposition of CLIL education that this type of pedagogical approach "should demonstrate some improvement in English [or other target] language learning and … present no long-term cost to academic content learning" (Macaro et al., 2018, p. 38). Therefore, based on the cost-benefit ratio for content and TL learning (Macaro, 2018), CLIL may be a worthwhile approach.

However, a closer look at the effect sizes in the forest plot revealed that our dataset had a relatively high degree of heterogeneity in effect sizes across the samples, suggesting the need for further investigation into the differences in research methodology (i.e., moderators) across

the identified studies.

The results of the moderator analyses revealed that homogeneity was a significant moderator, aligning with the meta-analyses of Lee et al. (2023) and Lo and Lo (2014). More specifically, the mean effect size was positive and significant (mean ES = 0.28) for those that did not confirm homogeneity between the CLIL and non-CLIL groups, whereas the mean ES was −0.16 for those that did. This finding suggests that the CLIL groups in studies that did not confirm the homogeneity of the groups may have consisted of students with higher levels of motivation and academic ability than those in mainstream groups. In short, the findings reported in the identified CLIL studies may not be exclusively due to CLIL instruction; rather, some CLIL programs may have been selective in terms of their students, with potential pre-existing differences between groups (Bruton, 2011a, 2011b). Although CLIL may not be as discriminatory as other educational programs (e.g., private schooling), as Huttner and Smit (2014) noted, some methodological issues seem to permeate CLIL research, as indicated by current and previous meta-analyses (Lee et al., 2023; Lo & Lo, 2014) on this issue.

However, Bruton's (2011a, 2011b), criticism of the selective nature of CLIL programs is being increasingly acknowledged in CLIL research based on quasi-experimental studies. Therefore, studies in this strand have started to adopt a randomized controlled trial (Piesche et al., 2016), administer a pre-test on target content knowledge (e.g., Dallinger et al., 2016; Fung, 2021), and employ a list of intelligence and aptitude tests to match the CLIL and non-CLIL groups (e.g., Madrid & Barrios, 2018; Martinez Agudo, 2022), with the latter sometimes involving the elimination of some outliers (Hughes & Madrid, 2020) to maximize the homogeneity of the CLIL and non-CLIL groups. Given these trends, future researchers should employ two or more different but complementary approaches to ensure homogeneity in the two groups as much as possible; this may reduce the possibility of overestimating the positive effects of the CLIL approach.

Another significant moderator found in the present meta-analysis was the nature of the content subjects, through which we categorized the subjects into those involving relatively more language elements (e.g., social studies and humanities) and those with fewer such elements (e.g., mathematics and science). Regarding this moderator, negative and positive mean effect sizes were found for the former and latter (−0.23 and 0.19, respectively), indicating that CLIL students' learning of content subjects may be less disadvantaged in subjects loaded with less language. This finding echoes that of Lo and Lo's (2014) meta-analysis of Hong Kong studies with secondary-level students, in which the performance of the CLIL group (EMI in their study) was significantly lower than its mainstream counterpart in the social science domain. This finding can be attributed to the different language registers in different disciplines. As reviewed above, in content subjects such as science and mathematics, teachers and students

are likely to draw on resources other than language (e.g., symbolic devices, manipulatives, and hands-on experiments) (O'Halloran, 1998), and these other forms of resources may reduce the language demands of students. In short, students' acquisition of content knowledge is less dependent on the language mode in such subjects, and they will be less disadvantaged when learning these subjects in their TL. By contrast, CLIL students' content learning was negatively affected in more language-demanding content subjects. This can be explained by the fact that the language associated with content subjects is usually academic language, which is different (and arguably more complicated) from everyday language with which students are familiar (Martin, 2013). These may, in turn, hinder students expressing their content learning in appropriate academic language in the TL, thereby affecting their content learning (Lo et al., 2023).

Given this finding, CLIL teachers are encouraged to provide additional support in content subjects that are relatively loaded with more language to provide more support for students in both the content and language dimensions, particularly the latter, as most CLIL or EMI teachers are content specialists. For instance, teachers can draw students' attention to academic language (e.g., academic vocabulary, grammar features, and sentence patterns) through language-focused episodes (An et al., 2019). Content teachers can also collaborate with language teachers to analyze their students' needs and map their curricula to design a more content- and language-integrated curriculum with explicit language objectives (e.g., academic language related to the content) and pedagogical approaches such as genre-based approaches (Cammarata, 2016; Lin, 2016). This calls for more professional development programs for CLIL teachers (Lo, 2020).

7. Conclusion

This meta-analysis contributes to the field of CLIL research by synthesizing the findings of existing primary studies on the effects of CLIL on secondary school students' acquisition of content knowledge in different pedagogical contexts. Moderator analyses showed that the previously reported superiority of the CLIL approach over mainstream approaches used in some primary studies may have been overestimated because of the lack of control for pre-existing differences between the CLIL and non-CLIL groups. Nonetheless, based on our findings, we propose that the CLIL approach could produce similar degrees of gain to that of its mainstream counterparts, suggesting no detrimental effects of CLIL instruction on secondary school students' content learning. Therefore, considering Macaro's (2018) cost-benefit ratio, this study's results and Lee et al.'s (2023) meta-analysis of the effects of CLIL on the development of TL proficiency collectively suggest a promising outlook for the CLIL approach. Notably, the present meta-analysis only considered studies in secondary school

contexts; thus, synthetic efforts concerning other learner populations, such as primary and tertiary students, are needed to confirm the aforementioned recommendations for CLIL instruction and to propose a more comprehensive evaluation of this bilingual approach. Nonetheless, the present meta-analysis takes an important step in this direction by adding weight to the pedagogical value of this approach.

It is worth mentioning some limitations in the present meta-analysis. First, our dataset included three samples from Yip et al. (2003) conducted in the Hong Kong context (one treatment and three comparison groups), with each group comprising more than 3000 participants, making it the largest study in the dataset. Although no publication bias was detected in our analysis (see *Outlier diagnosis and publication bias* section for details), this study may have influenced the overall results owing to its large sample size. This calls for future research with larger sample sizes in other educational contexts, allowing meta-analysts to obtain more balanced and comprehensive results. Second, we did not include the 18 identified one-group studies (i.e., within-group design) in our analysis because this design cannot adequately rule out alternative explanations for changes (e.g., maturation effects, practice effects) when estimating treatment effects (Cook & Campbell, 1979; Dörnyei, 2007). While including these studies would have increased the sample size, it would have diverted us from our primary research objectives. Moreover, based on our preliminary analysis, their inclusion would not have helped identify additional moderator variables. Instead, we believe that the lack of detailed methodological descriptions in the included studies was a key factor limiting the identification of certain important moderator variables. These variables may include the characteristics of CLIL teachers (e.g., their years of CLIL teaching, proficiency in the TL), the intensity of CLIL programs, and configurations of CLIL lessons, to list just a few. Therefore, CLIL researchers should provide more detailed descriptions of their methodology, allowing future meta-analyses to consider a wider range of potential moderators.

Finally, we suggest the following directions for future research on the effects of CLIL on content knowledge development. First, more research is needed on the expertise and majors of CLIL teachers. Since teacher instruction is an essential component of CLIL implementation, the effects of CLIL teachers' backgrounds on content knowledge deserve more attention. Second, the development of more standardized instruments to measure students' content knowledge would be beneficial for research on the effects of the CLIL approach, as extant primary studies have used a wide range of measures that differ in format, scope, and content, making it difficult to conduct a systematic investigation of the effects of this approach. Finally, we call for the establishment of a global standard for determining the intensity of CLIL

programs. Such a standard would be useful not only for future meta-analysts to include this moderator in their analyses, but also for CLIL program administrators to determine the intensity of their programs based on meta-analysis results.

CRediT authorship contribution statement

Fatima Kaiypova: Writing – original draft, Data curation, Conceptualization. **Hansol Lee:** Writing – review & editing, Writing – original draft, Software, Methodology. **Yuen Yi Lo:** Writing – review & editing, Writing – original draft, Data curation. **Jang Ho Lee:** Writing – review & editing, Writing – original draft, Supervision, Software, Data curation, Conceptualization.

Data availability statement

Data will be made available upon request.

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Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.system.2024.103580.

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